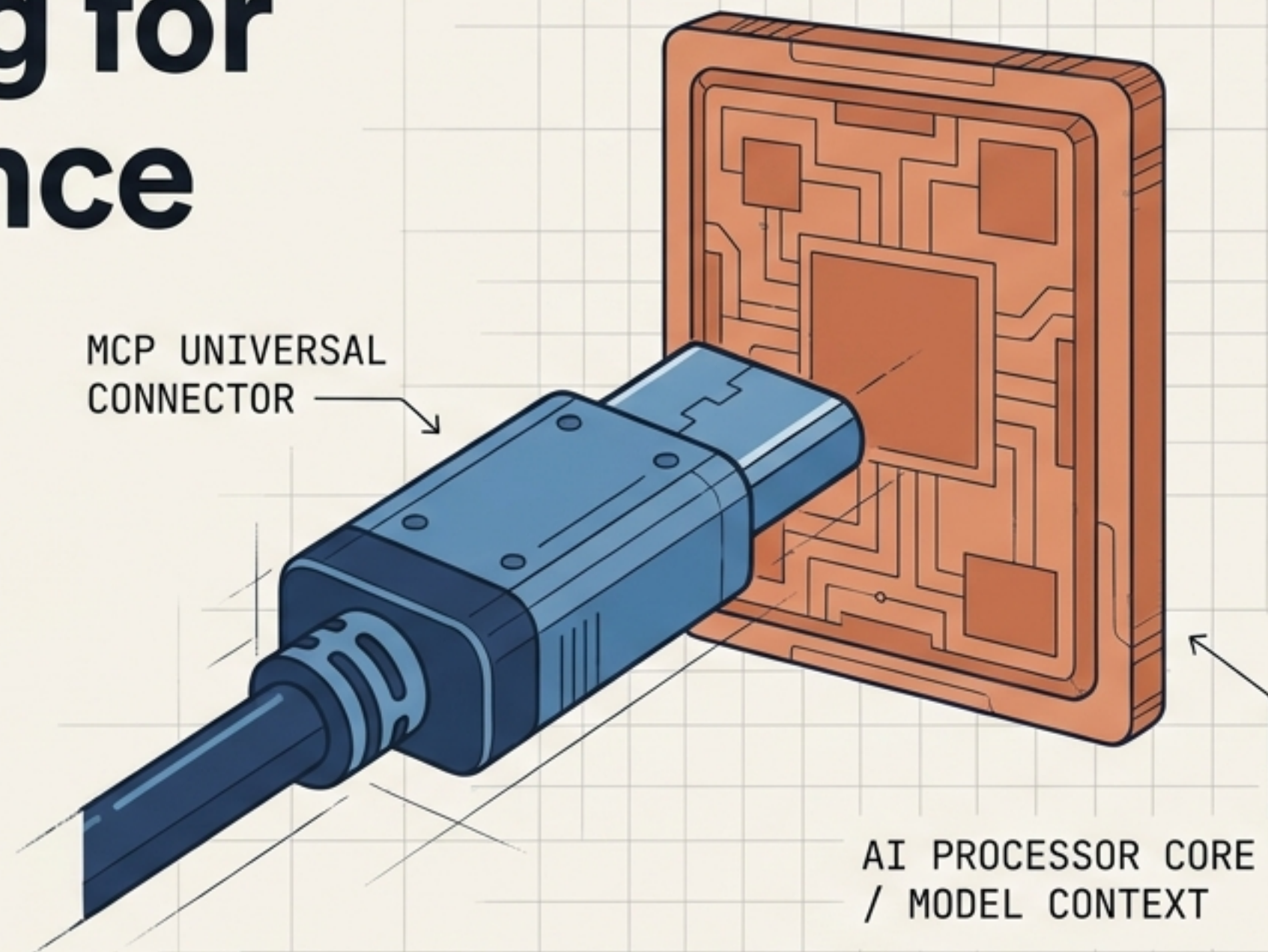


The Universal Plug for Artificial Intelligence

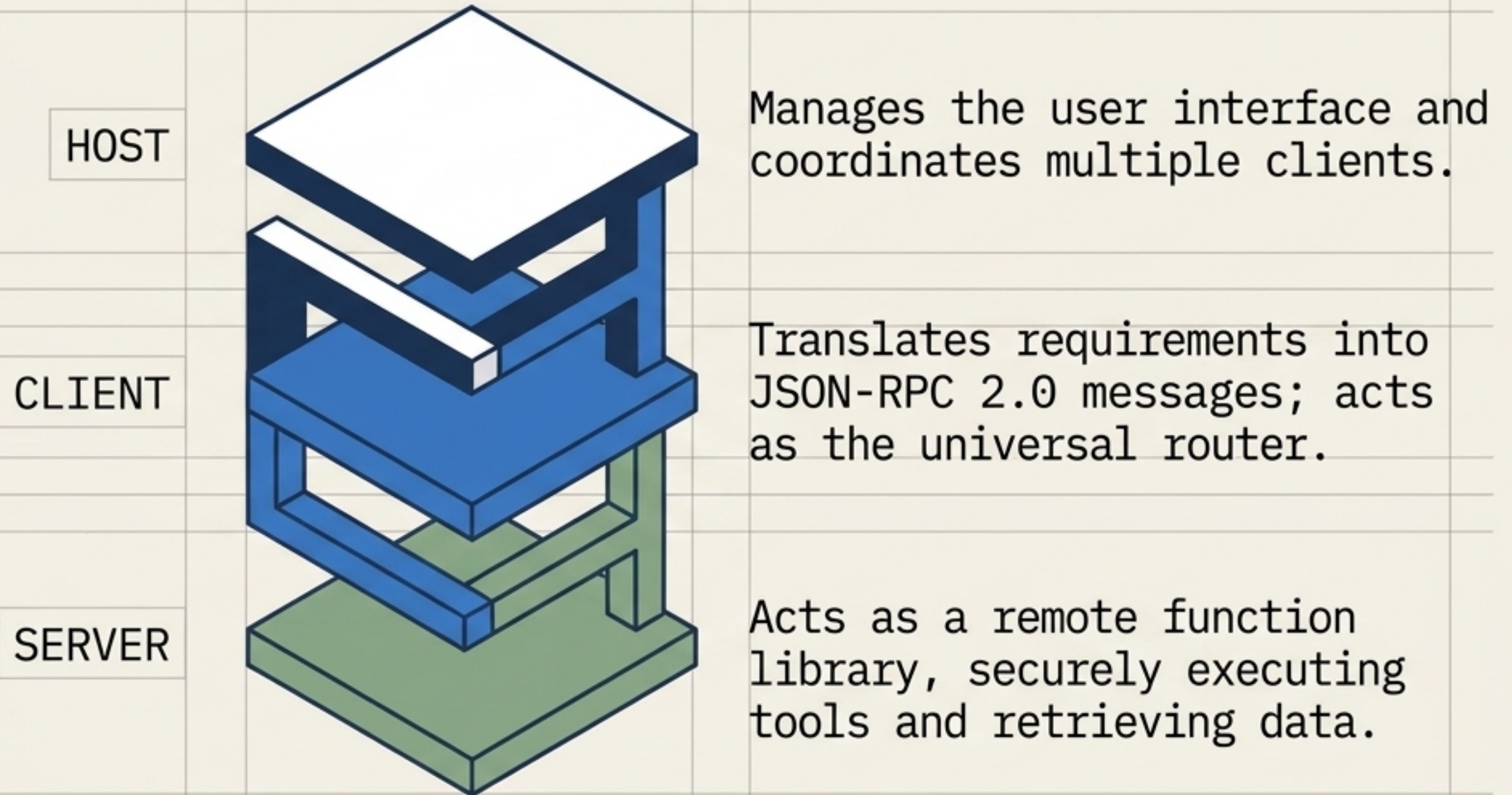
- No more bespoke API integrations.
- No more fragmented agentic workflows.
- The Model Context Protocol (MCP) standardizes how AI connects to the real world.



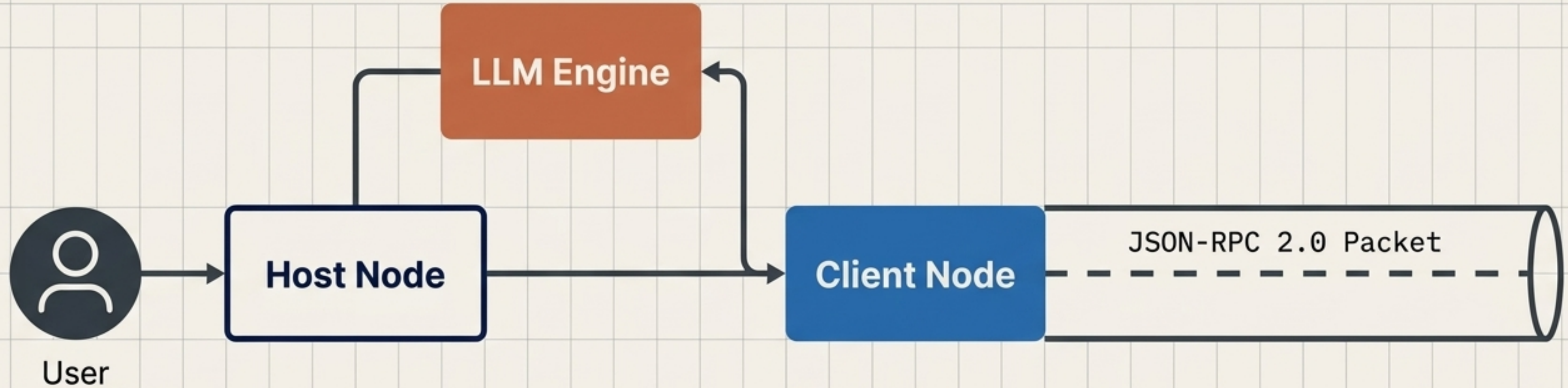
Moving from static integration to dynamic discovery

	Traditional APIs	MCP (Model Context Protocol)
Endpoints	Fixed and predefined	Dynamically discovered at runtime
Integration	Requires custom connector code	Plug-and-play via a universal protocol
Scope	Transfers raw data only	Transfers data, executable tools, and LLM prompts
Interaction	Static request/response	Context-aware, bidirectional, agentic workflows

A three-layer architecture separating concerns



The Client acts as a universal router, not an executor



Session Management

Handles lifecycles, interruptions, and clean terminations.

Format Translation

User intent is strictly formatted into standard JSON-RPC.

Pure Orchestration

Routes the LLM's requests perfectly without executing the tasks itself.

Servers act as protective shells around enterprise systems

Agentic Support

Keeps data fresh and minimizes hallucinations by providing real-time ground truth to the LLM.



Action Translation

Servers convert MCP JSON-RPC requests into secure, localized system actions.

Capability Exposure

Dynamically advertises executable tools via machine-readable catalogs.

Three standardized buckets for all AI capabilities



TOOLS

(Active)

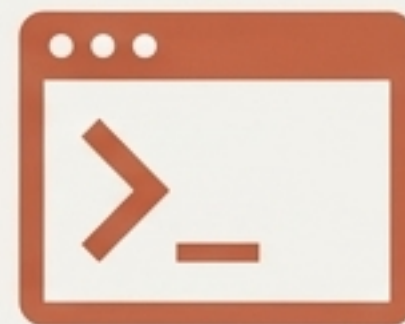
Executable functions that cause state changes (e.g., query database, call an API). AI explicitly decides when to invoke them.



RESOURCES

(Passive)

Read-only contextual data accessed via URI templates. AI passively reads this data to gain ground-truth context.

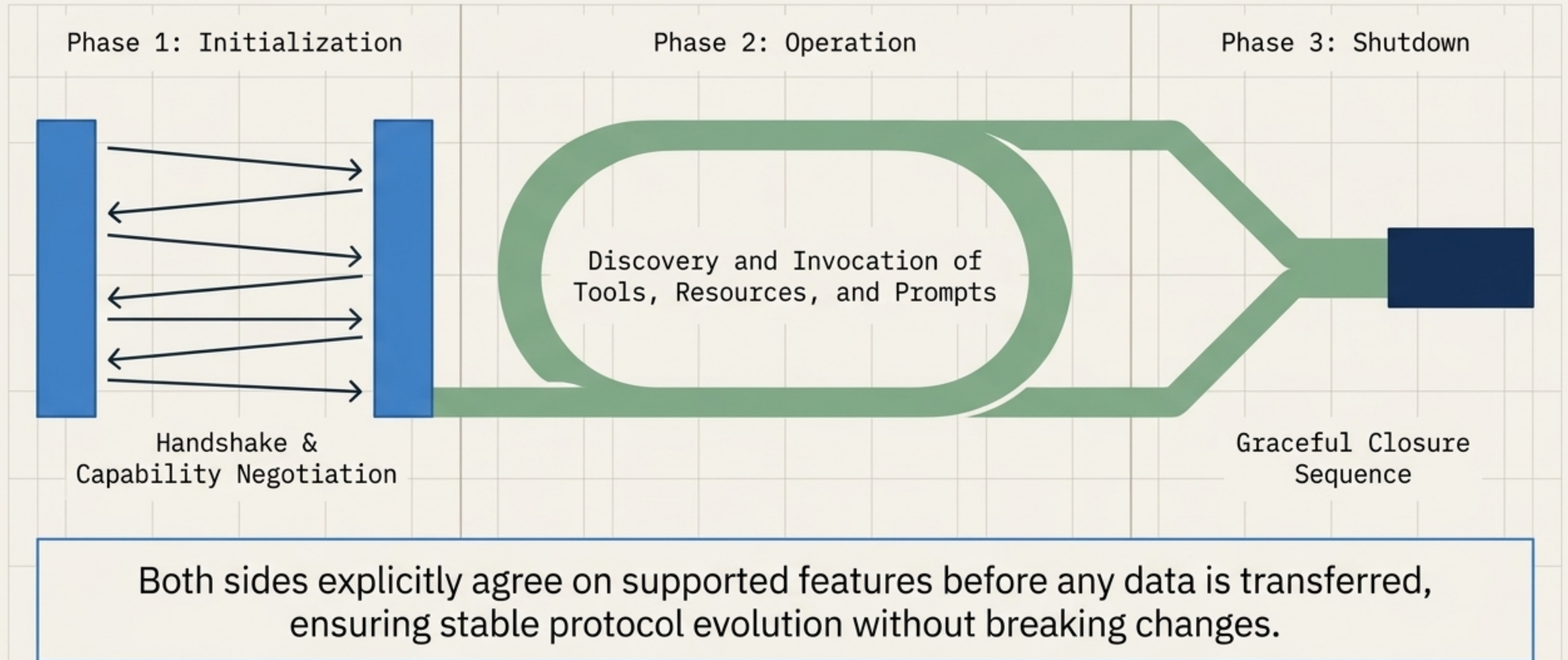


PROMPTS

(Reusable)

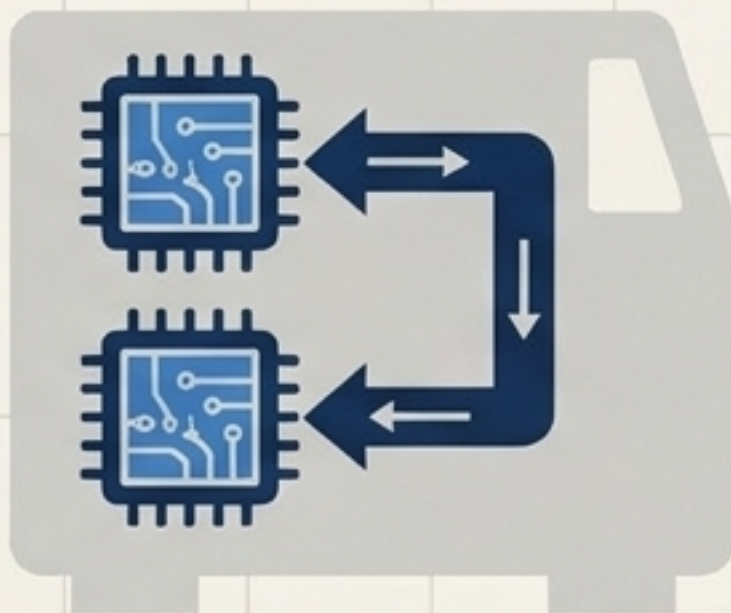
Pre-defined instruction templates and workflows that structure LLM interactions for guaranteed consistency.

A negotiated lifecycle ensures graceful capability degradation



Adapting pathways without altering application logic

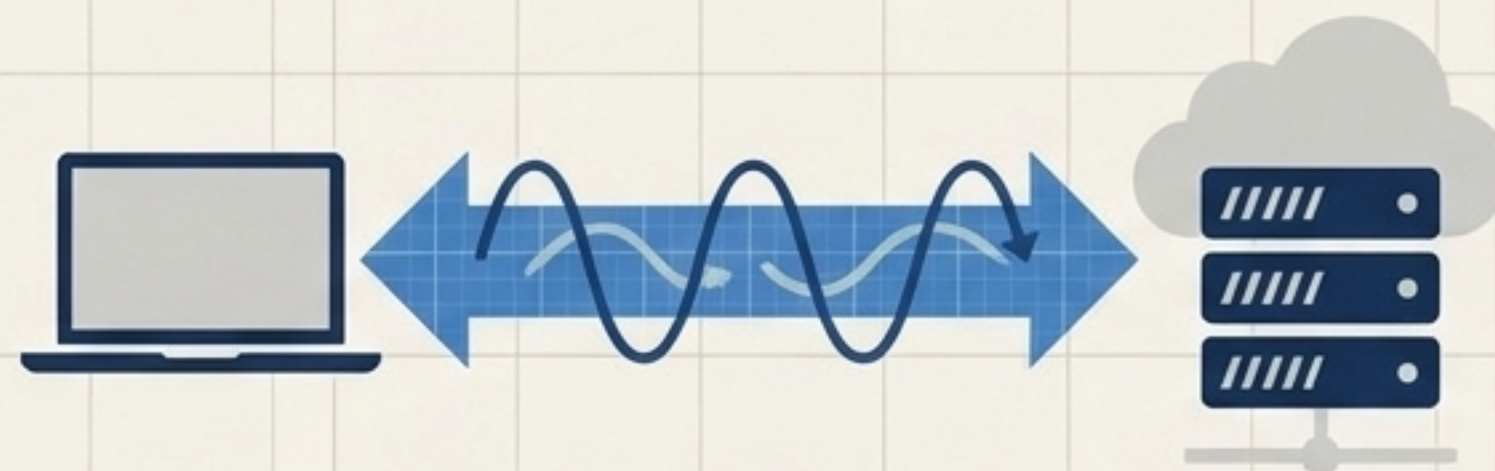
STDIO Transport



Local, process-bound, highly secure standard input/output streams.

Ideal for local testing and secure file system integration.

Streamable HTTP

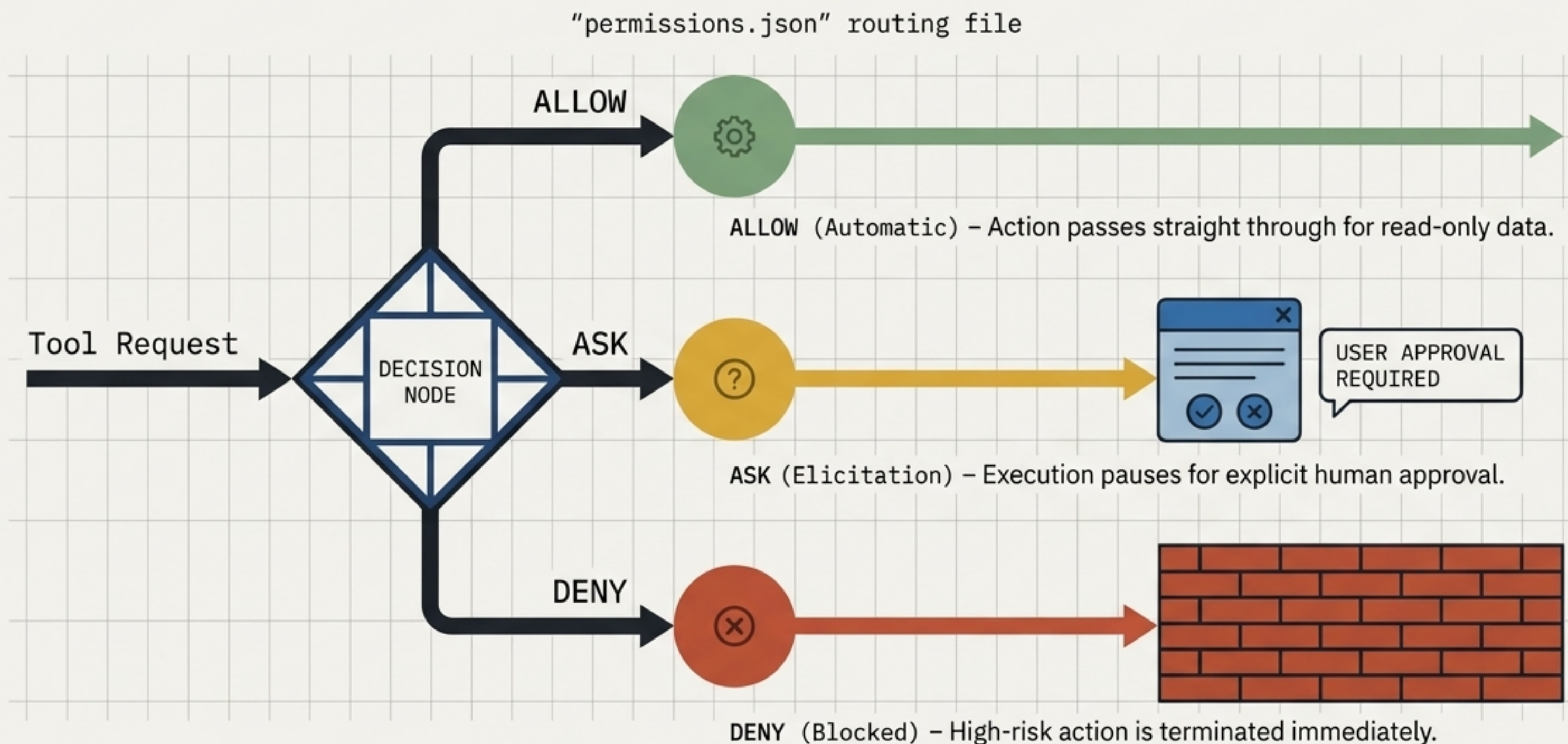


Remote, bidirectional HTTP connections (replacing legacy SSE).

Built for cloud servers, multi-client scaling, and robust error recovery.

Zero logic changes required to switch between them.

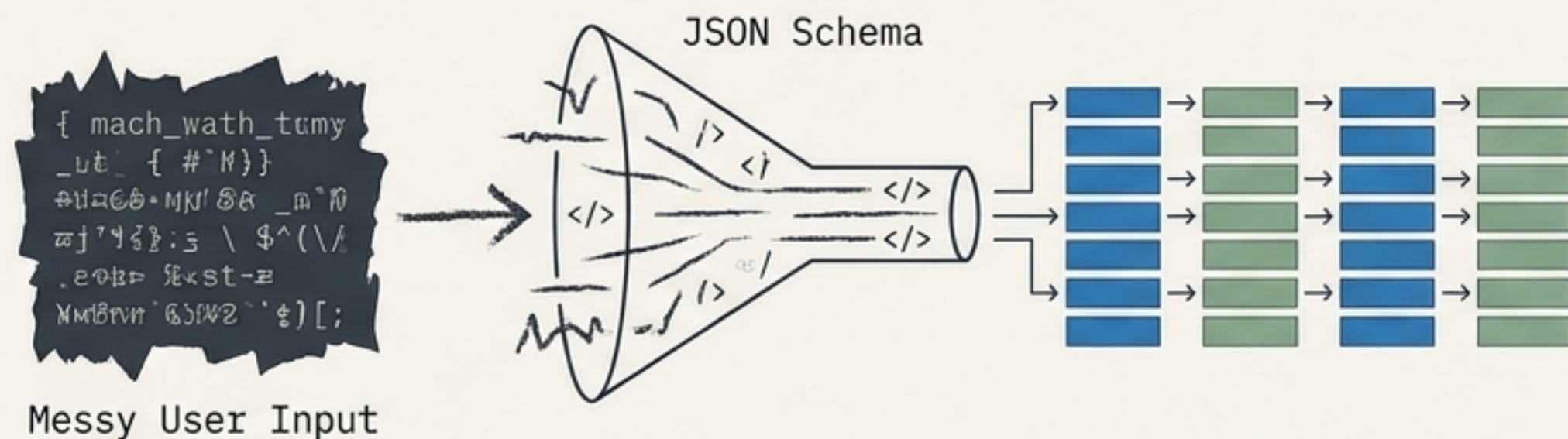
Security begins before a message ever leaves the client



Client-Side Enforcement: Least-privilege policies protect users from autonomous agent mistakes locally.

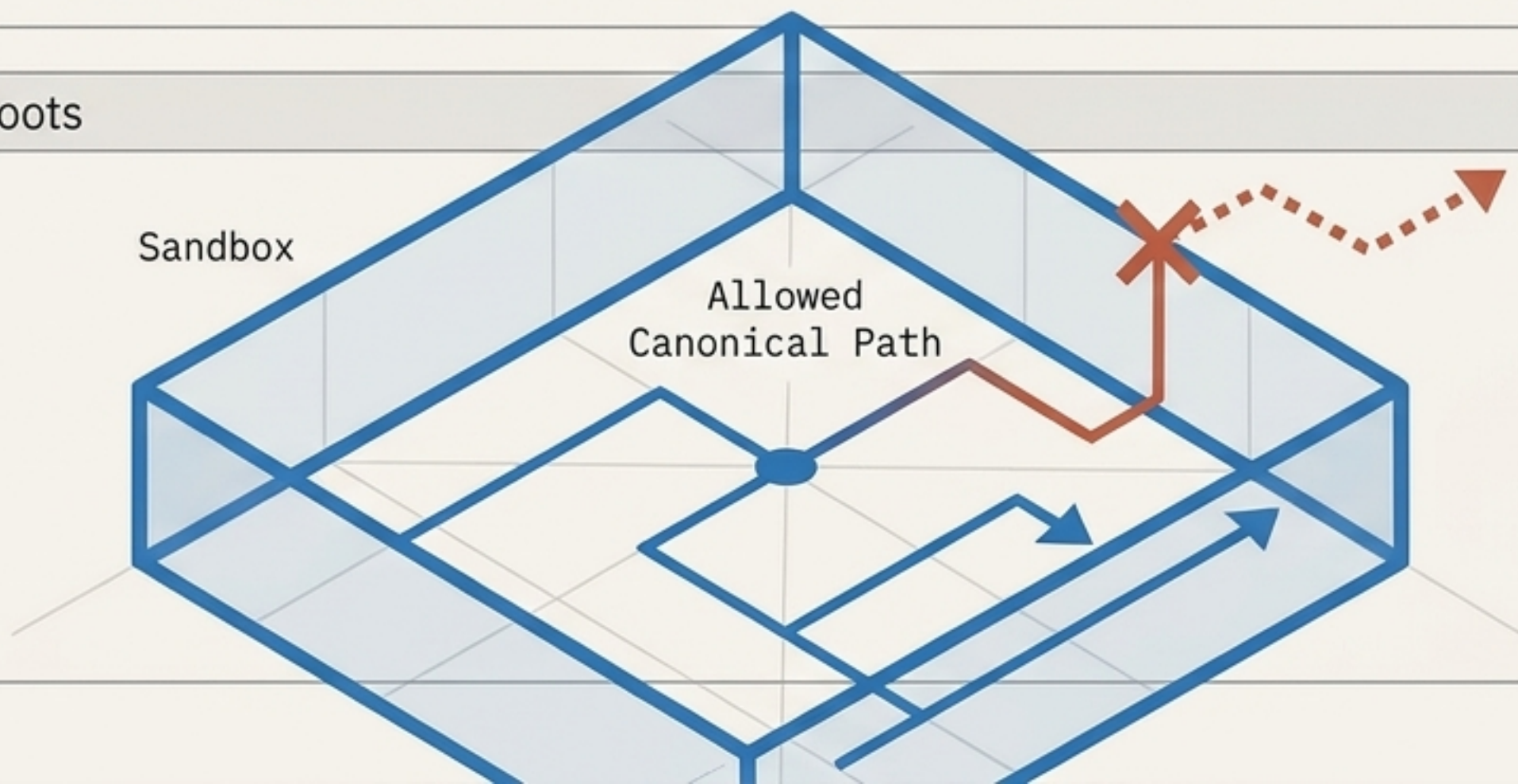
Bounding the sandbox and validating the intent

Elicitation



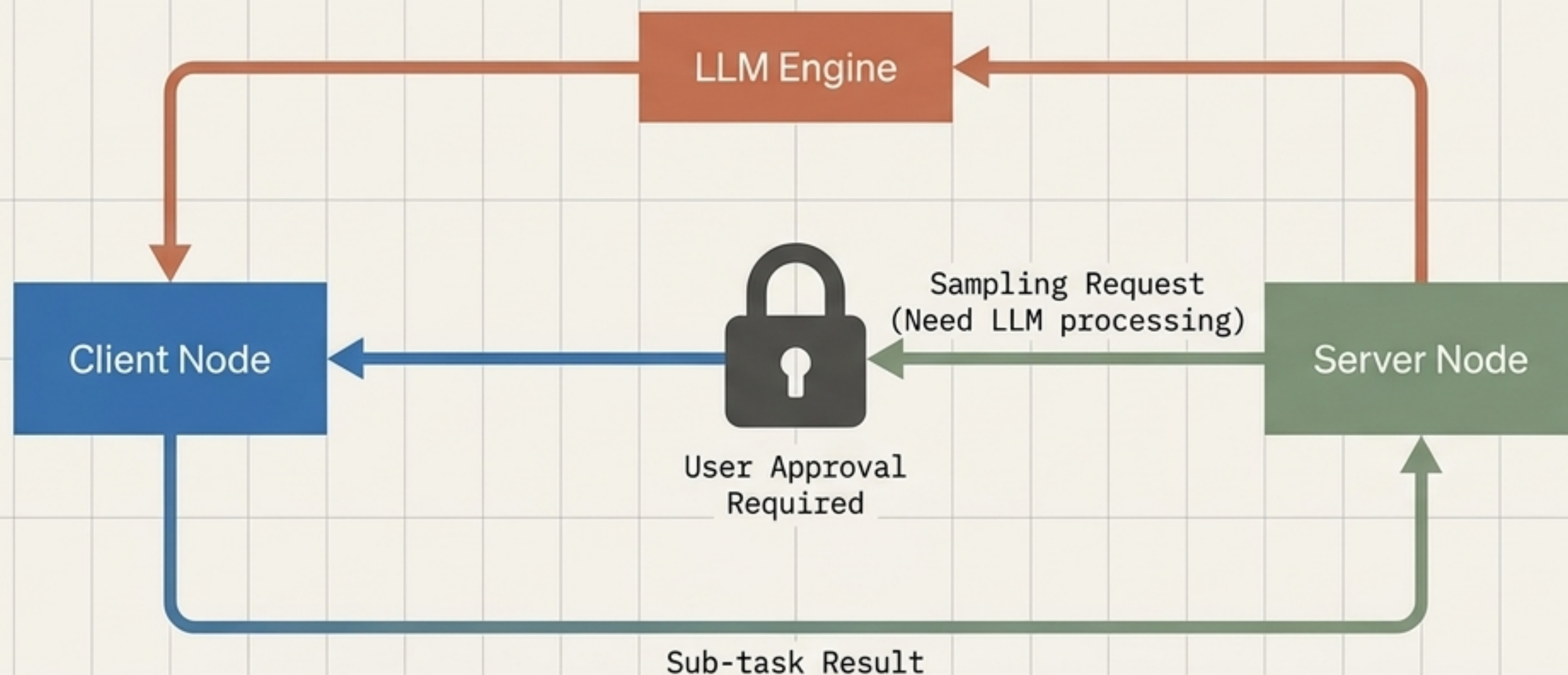
Server-initiated structured input validation ensures informed user consent and perfectly formed execution data.

Roots



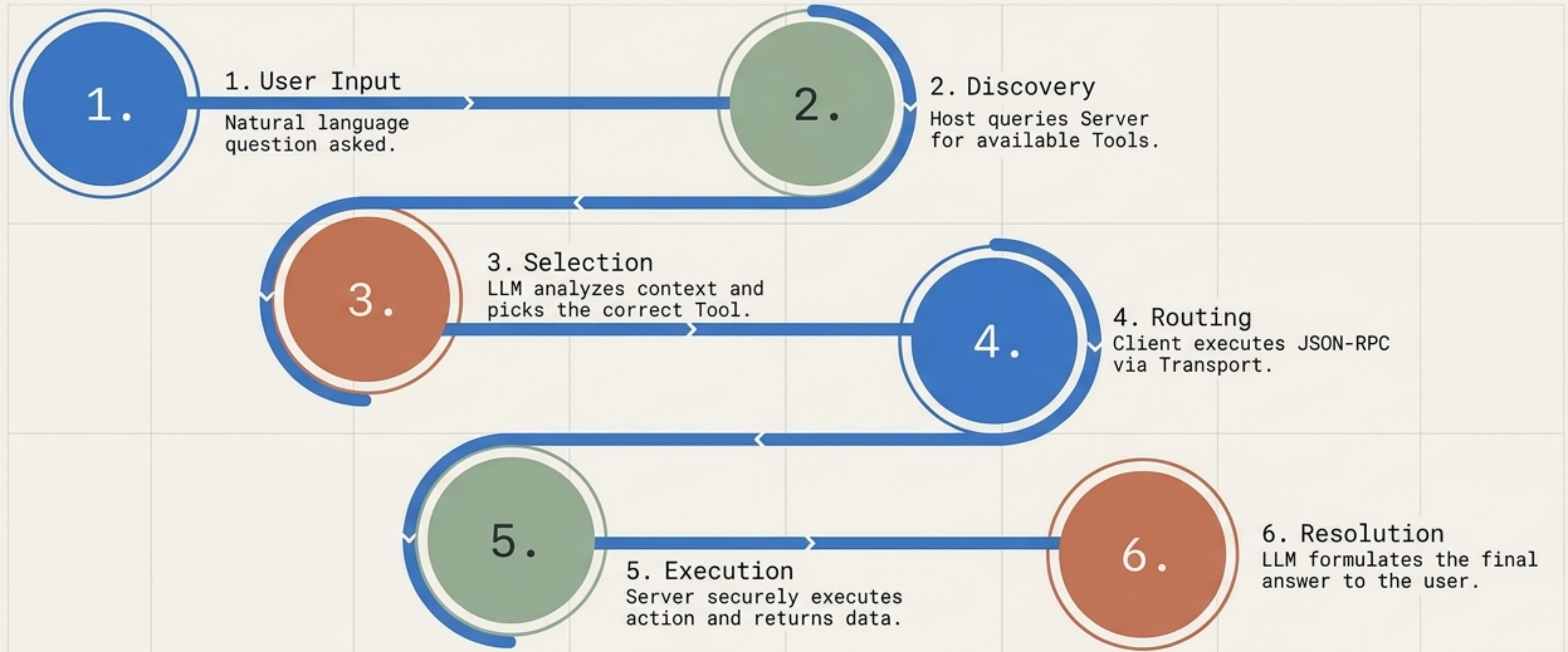
Strict boundaries force servers to resolve canonical paths, denying any operation attempting to access data outside defined directories.

Sampling: Securing complex, multi-step LLM operations

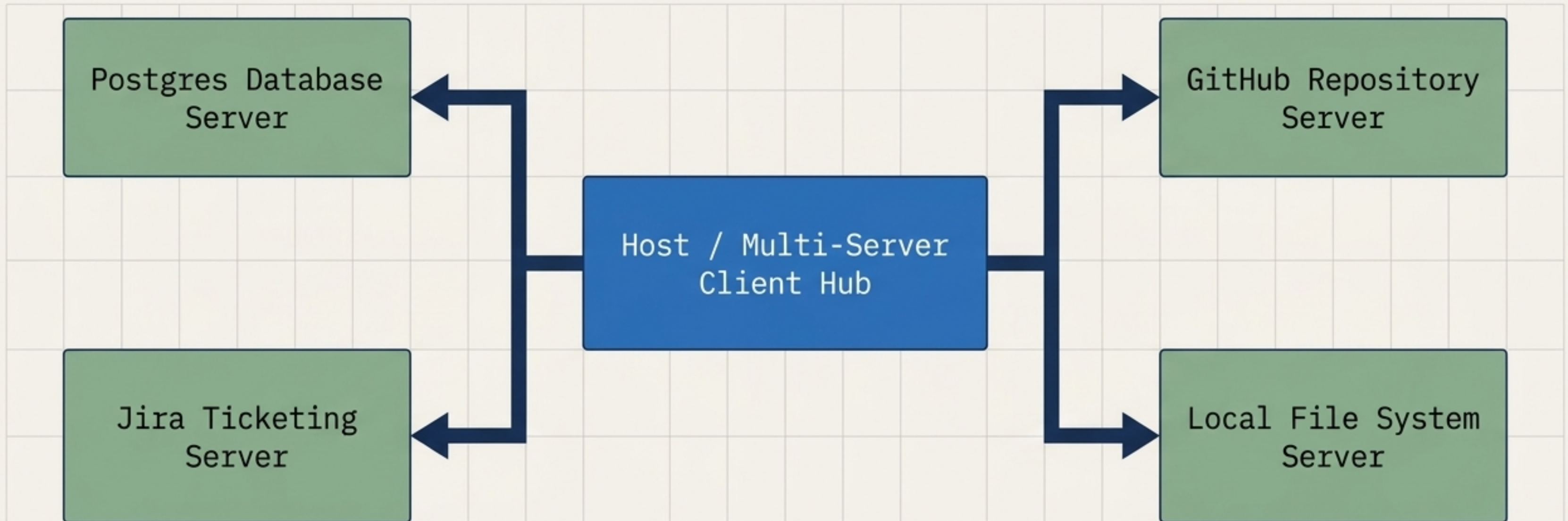


Servers can request advanced LLM capabilities from the client to complete complex sub-tasks. Crucially, this is always gated by **explicit user consent**, preventing runaway autonomous chains.

Anatomy of a request: From natural language to real-world action

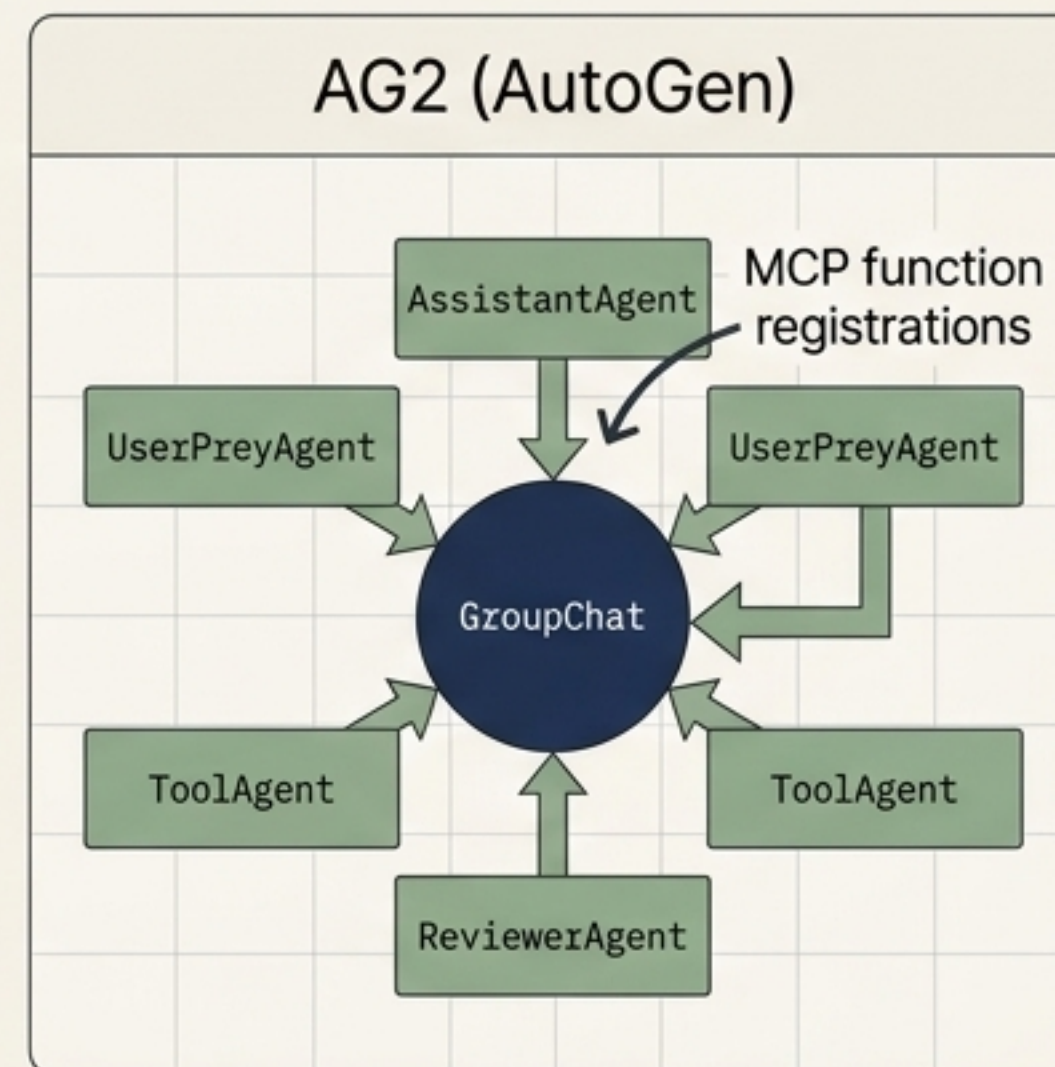
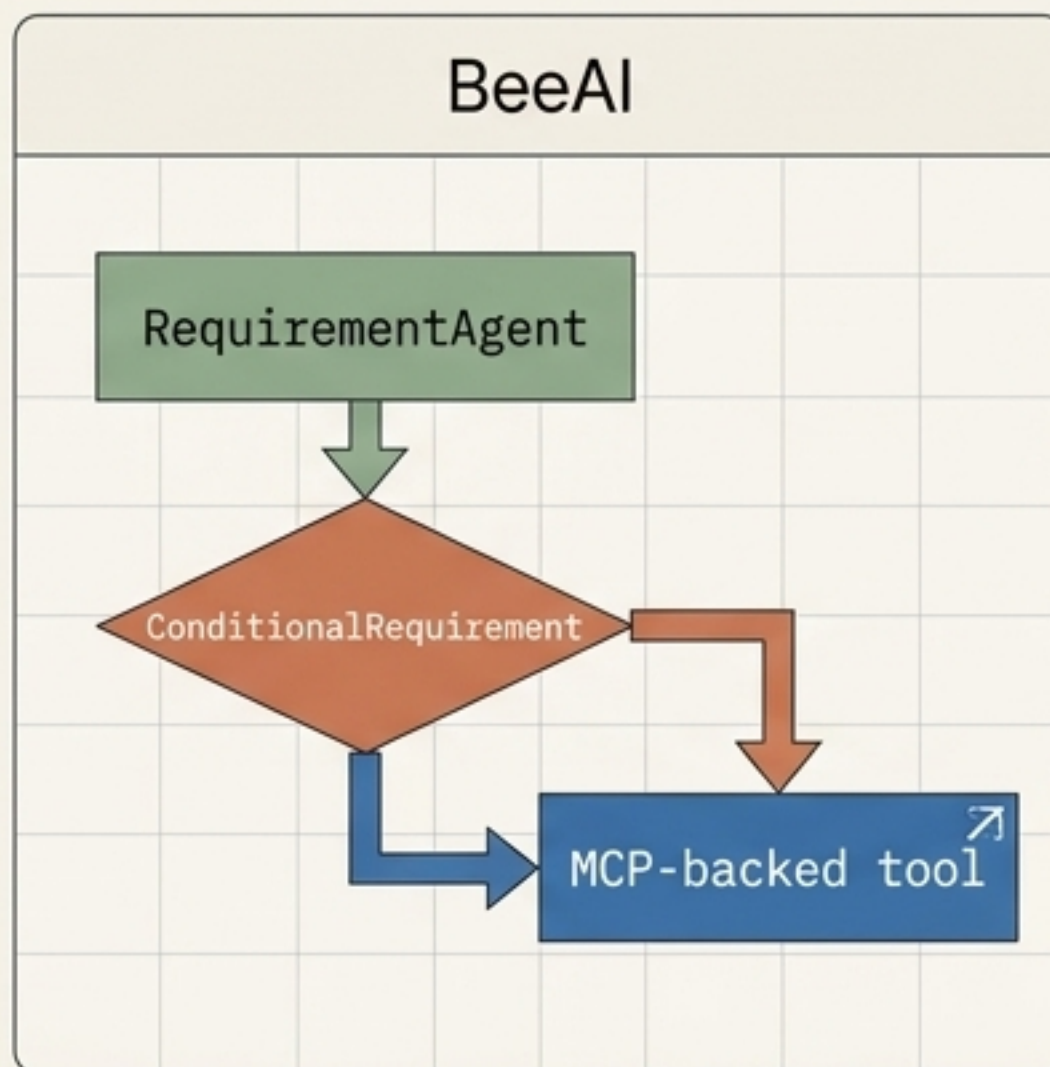
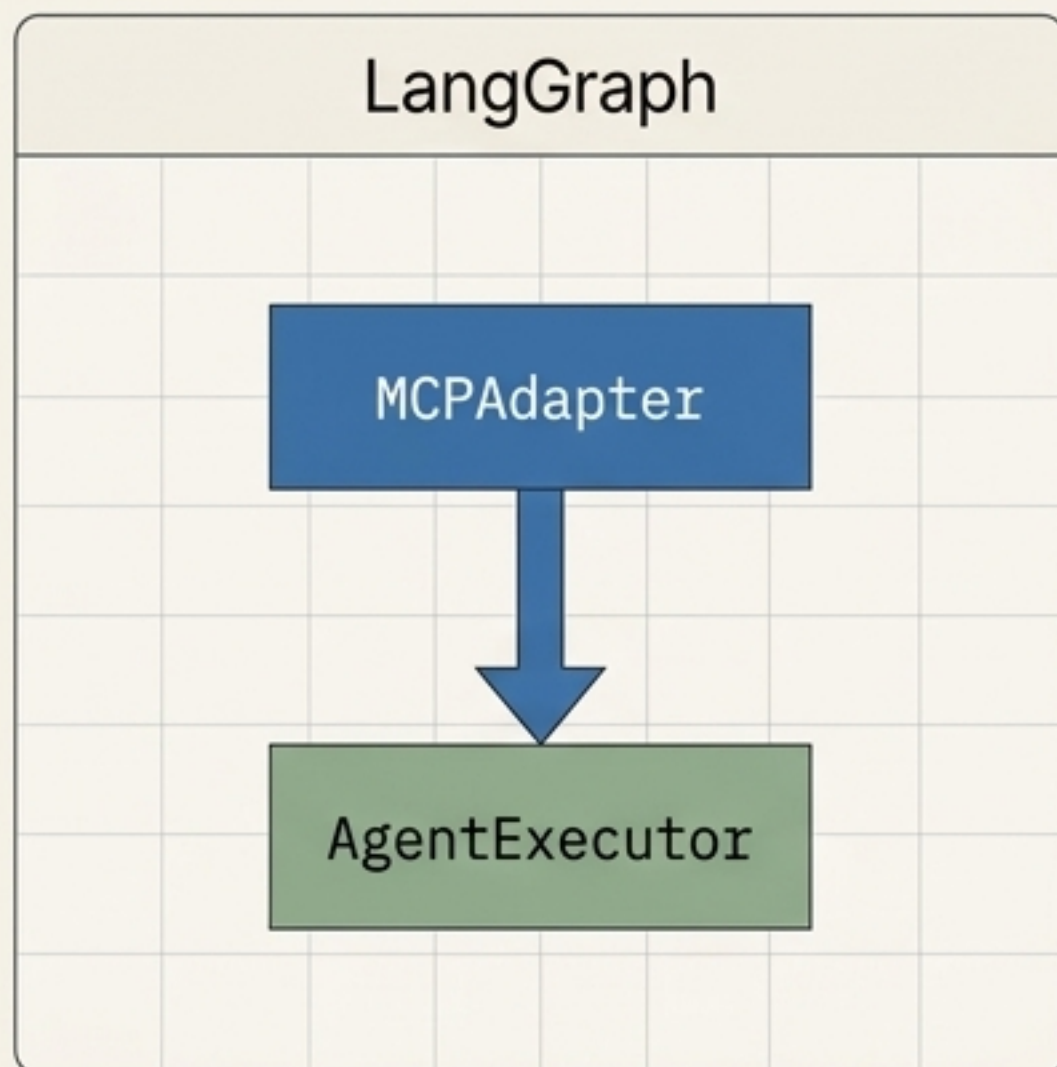


Infinite domain orchestration without custom code



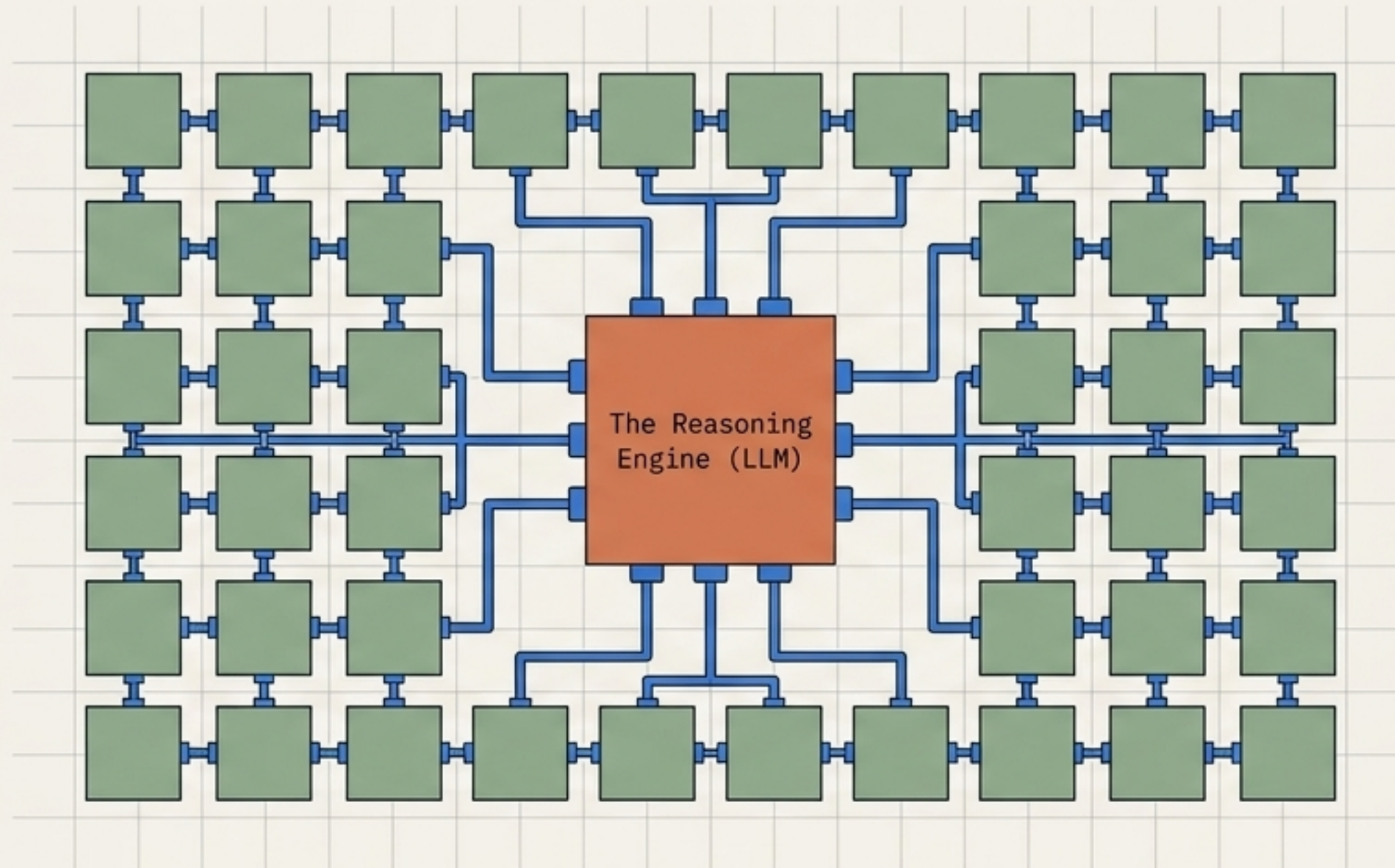
- The Host aggregates all tools into a unified array for the LLM.
- An AI agent can query a database, write a code fix, and update a ticket simultaneously.

Native integration with leading agentic frameworks



MCP doesn't replace agent frameworks; it supercharges them. Frameworks instantly adapt MCP server capabilities into their native tool structures, drastically reducing time-to-market.

The separation of reasoning and capability



By entirely decoupling the AI reasoning engine from the underlying external systems, MCP ends the era of fragmented AI integrations. We unlock truly autonomous, infinitely scalable, and universally secure AI applications.